DAY 11 – NETWORKING FUNDAMENTALS

1. INTRODUCTION

Computer networking refers to connecting two or more devices to exchange information, share resources, and communicate with each other.

Networks are widely used in organizations to connect computers, servers, printers, and other devices.

Understanding networking fundamentals is essential in cybersecurity because attackers often exploit weaknesses in network configurations, protocols, and connected devices.

The important networking concepts include:

- IP addresses
- MAC addresses
- Network ports
- Communication protocols
- Routers
- Switches
- Firewalls

2. OBJECTIVE

The main objective is to understand basic networking concepts and their importance in cybersecurity.

Specific Objectives

1. Understand IP and MAC addressing.
2. Learn the purpose of network ports.
3. Study common networking protocols.
4. Understand the functions of routers, switches, and firewalls.
5. Design a simple network topology using Cisco Packet Tracer.
6. Understand how devices communicate within a network.
7. Learn basic network connectivity testing.

3. IP ADDRESS

An Internet Protocol (IP) address is a logical address assigned to a device connected to a network. It helps identify devices and route data between networks.

Types of IP Addresses:

IPv4: A 32-bit address written in dotted-decimal format.

Example: 192.168.1.10

IPv6: A 128-bit address designed to support a much larger address space.

Example: 2001:db8::1

Private IP Address: Used within local networks, such as 192.168.1.10.

Public IP Address: Used for communication over the public Internet.

Static IP: Manually assigned or consistently reserved.

Dynamic IP: Automatically assigned, commonly using DHCP.

4. MAC ADDRESS

A Media Access Control (MAC) address is a link-layer address associated with a network interface. Ethernet MAC addresses are commonly 48 bits long.

Example: 00:1A:2B:3C:4D:5E

MAC addresses help devices communicate within a local network. Ethernet switches use learned MAC addresses to forward frames toward the appropriate destination.

IP Address vs. MAC Address

IP Address	MAC Address
Logical network-layer address	Link-layer address
Used for routing across networks	Used for local network frame delivery
IPv4 or IPv6 format	Commonly 48-bit hexadecimal format
Can change with network configuration	May be hardware-assigned or software-configured

5. NETWORK PORTS

A network port is a logical number used by transport protocols such as TCP and UDP to identify an application or service.

Port numbers range from 0 to 65535.

Port	Protocol/Service	Purpose
20/21	FTP	File transfer
22	SSH	Secure remote access
25	SMTP	Email transfer
53	DNS	Domain name resolution
67/68	DHCP	Automatic IP configuration
80	HTTP	Web communication
110	POP3	Email retrieval
143	IMAP	Email access
443	HTTPS	Encrypted web communication

6. NETWORK PROTOCOLS

A network protocol is a set of rules governing communication between devices.

TCP (Transmission Control Protocol): Provides reliable, ordered data delivery.

UDP (User Datagram Protocol): Provides connectionless datagram delivery without built-in guarantees of reliability or ordering.

HTTP: Transfers web content.

HTTPS: Uses TLS to protect web communication.

DNS: Translates domain names into IP addresses and provides other naming records.

DHCP: Automatically assigns network configuration information.

ICMP: Supports network diagnostics and error reporting, including ping.

7. NETWORK DEVICES

Router

A router connects different IP networks and forwards packets between them.

Switch

A switch connects devices within a local area network and forwards Ethernet frames using MAC addresses.

Firewall

A firewall monitors and controls network traffic according to configured security rules.

Hub

A hub repeats received signals to other connected ports without selectively forwarding traffic by destination MAC address.

Wireless Access Point

A wireless access point allows wireless devices to connect to a network.

Router vs. Switch vs. Firewall

Device	Main Function
Router	Connects different networks
Switch	Connects devices within a LAN
Firewall	Enforces network traffic security rules

8. PRACTICAL IMPLEMENTATION USING CISCO PACKET TRACER

Cisco Packet Tracer is a network simulation and visualization application developed by Cisco. It allows students to build network topologies, configure devices, and observe packet movement without physical networking hardware.

9. IMPORTANCE OF NETWORKING IN CYBERSECURITY

Networking knowledge helps cybersecurity professionals:

1. Identify unauthorized devices and suspicious connections.
2. Understand network traffic and communication protocols.
3. Configure routers, switches, and firewalls securely.
4. Detect unusual network activity.
5. Investigate network-based attacks.
6. Understand IP addresses, ports, and services involved in security incidents.

10. CONCLUSION

Networking fundamentals are essential for understanding cybersecurity. IP addresses, MAC addresses, ports, protocols, routers, switches, and firewalls all play important roles in network communication and protection.

Cisco Packet Tracer provides a virtual environment for practicing network design, device configuration, and connectivity testing.

The proposed experiment demonstrates how two computers can communicate through a switch using static IP addresses. Completing the simulation and verifying connectivity will strengthen practical networking knowledge required for cybersecurity.

For your Word report, add three screenshots from your own Packet Tracer experiment: the completed network topology, IP configuration, and successful ping result. This will make the submission more practical and directly connected to Cisco's official learning activities.